

HTML5 Semantics + CSS Essentials



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Session Goals



SESSION GOALS

By the end of this session,
you will be able to

- Use HTML5 semantic tags properly to structure web pages
- Organize Code & content using semantic tags
- Understand how different display values affect the layout
- Learn how to position elements using position and z-index
- Build a basic multi-section web page layout using semantic tags and CSS

S01

Topics Covered



Topics Covered

These are the main topics we'll cover today to help you build your first web page

01. What is HTML5?

- HTML5 Document
- Common Semantic Tags (<header>, <nav>, <main>, etc.)
- Semantic vs Non-Semantic Tags
- bdi tag
- Audio & Video
- New Form Elements
- Building Structure with Meaning

02. CSS Essentials

- Styling elements on mouse interaction
- Making elements semi-transparent
- Adding images as backgrounds
- Handling content that overflows its box
- Display Elements
- Position of the elements

03. Hands-On Tasks

- **Task 1** : Task: Media Integration - Audio & Video
- **Task 2**: Hover & Opacity Interaction
- **Task 3** : Build a small blog-style layout using semantic HTML and basic CSS layout concepts

What is HTML5?



WHAT IS HTML5?

What is HTML5?

HTML5 is the latest version of the HyperText Markup Language used to structure and present content on the web.

HTML5 helps developers write cleaner, more meaningful, and more accessible code, and replaces old hacks with proper tags.

*"Think of **HTML5** like a **software update** for your phone.*

*Your phone is still the **same device**, but the new update **adds features, improves performance, and fixes old issues**.*

***HTML5** didn't replace **HTML** — it improved it."*

HTML5

Document Structure



HTML5 Document Structure

Every **HTML5** page follows a standard structure that helps the browser understand and render the content correctly.

<!DOCTYPE html> : Declares that this page uses HTML5,
It helps the browser understand how to render the page correctly

Semantic Tags



What are Semantic Tags?

Semantic HTML means using tags that clearly describe the meaning and purpose of the content they contain.

Why Use Semantic Tags?

1. *Easier to read and understand (for humans and developers)*
2. *Helps search engines (**SEO**) understand your content*

"Before HTML5, we had to use <div> for everything — it was like calling everyone 'dude'.

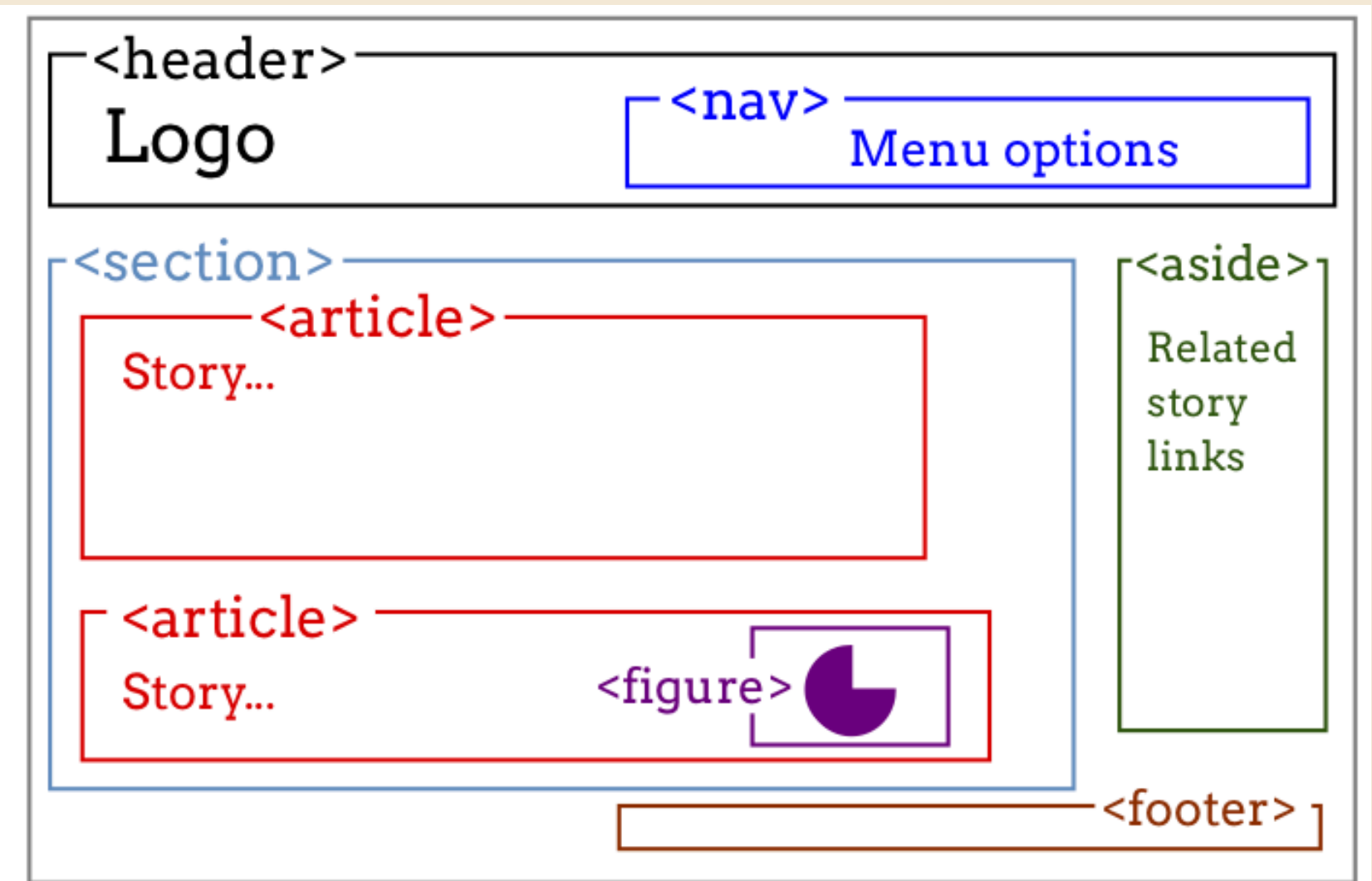
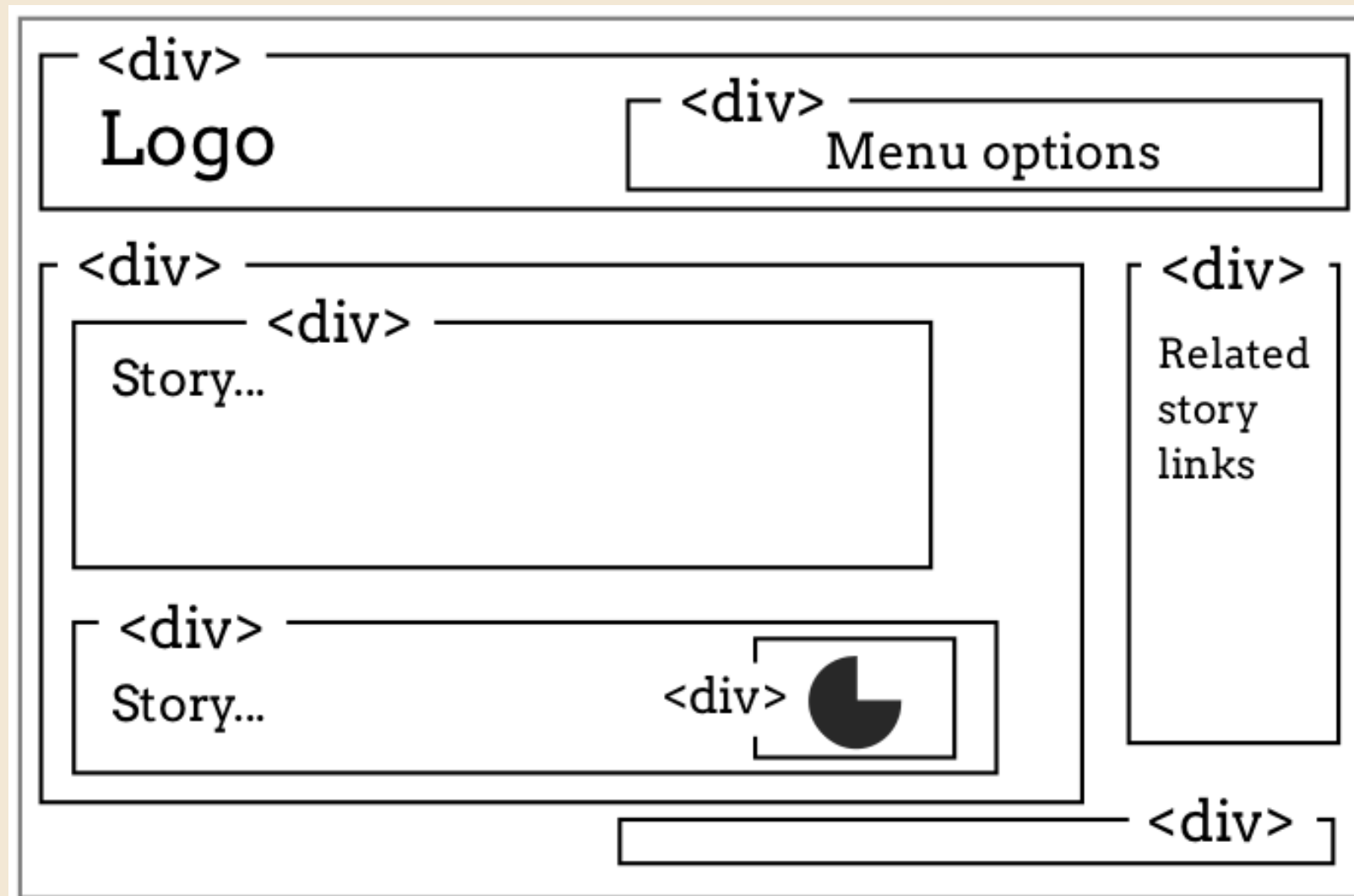
Now with semantic tags, it's like we finally know everyone's name."

What are Semantic Tags?

Semantic Tag :

1. **<header>** : Used for introductory content at the top of a page or section (e.g. page header + article header)
2. **<nav>** : Contains navigation links, Helps screen readers and search engines understand the site structure
3. **<main>** : Wraps the main content of the page, It's the "meat" of your webpage — the reason people are here
4. **<section>** : Groups related content together, Each section should have a heading
5. **<article>** : Represents independent, reusable content, Could be a blog post, news article, forum post, etc.
6. **<aside>** : Contains side content related to the main content
7. **<footer>** : Appears at the bottom of a page or section, Contains copyright, contact info, navigation, or social media
8. **<figure>** : Used to wrap self-contained media content, like: images, diagrams, illustrations, charts, or even code blocks
9. **<figcaption>** : A caption or description for the content inside <figure>

What are Semantic Tags?



<bdi>

Bi-Directional Isolation



<BDI> – BI-DIRECTIONAL ISOLATION

What it does:

<bdi> stands for Bi-Directional Isolation :

used to prevent text direction issues when mixing left-to-right (LTR) and right-to-left (RTL) languages (e.g English & Arabic).

User: <bdi>علي</bdi> - Score: 90

User: <bdi>John</bdi> - Score: 85

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Mini Task



MINI TASK : LET'S TRY IT NOW! — YOU HAVE 10 MINUTES. (RAISE YOUR HAND IF YOU NEED HELP.)

Mini Task: Media Integration — Audio & Video

Objective:

*Practice how to embed and control **audio and video** elements using **HTML5**.*

Requirements :

1. *Open **VS Code***
2. *Create a new HTML page called **media-task.html***
3. *Add the standard **HTML structure***
4. *Add an `<audio>` element*
5. *Add a `<video>` element*

S01

HTML5 Forms



New HTML5 Form Elements

HTML5 introduced several new input types that enhance user experience and improve form validation.

Types :

- *<input type= "**color**"> : Allows users to pick a color using a color picker*
- *<input type= "**range**"> : Renders a slider to select a number in a range, You can set min, max, and step*
- *<input type= "**search**"> : Optimized for search inputs, Some browsers may show an "×" to clear the text*
- *<input type= "**url**"> : Expects a valid URL format (e.g., https://site.com)*
- *<input type= "**time**"> : Lets users choose a specific time (e.g., 10:30 AM)*
- *<input type= "**date**"> : Displays a calendar picker*

S02

CSS Essentials



CSS Essentials

:hover : A pseudo-class used to apply styles when the user hovers the mouse over an element, Used to create interactive effects like buttons changing color.

```
button:hover {  
    background-color: blue;  
    color: white;  
}
```

opacity : Controls how transparent an element is, 1 = fully visible, 0 = fully transparent

```
img {  
    opacity: 0.5;  
}
```


CSS Essentials

background-image : property is used to set an image as the background of an element (e.g., div, section, body, etc.)

```
selector {  
  background-image: url("image.jpg");  
}
```

Related Properties:

- **background-repeat** : Controls if/how the image repeats (**repeat**, **no-repeat**, **repeat-x**)
- **background-size** : Resizes the image (**auto**, **cover**, **contain**)
- **background-position** : Controls the position (**top**, **center**, **right**, **etc.**)
- **background-attachment** : Controls scroll behavior (**scroll**, **fixed**)

CSS Essentials

display : Controls how an element is rendered in the layout.

```
p {  
  display: inline;  
}
```

Main values :

- **block** : *Takes full width of the container and always starts on a new line.*
- **inline** : *Sits beside other elements on the same line, width and height cannot be set.*
- **inline-block** : *Behaves like inline (stays in line) but allows you to set width and height Also, margin and padding work fully on all sides*
- **none** : *Controls how scrolling works (e.g., scroll, fixed, none), used to show/hide elements or fix position during scroll.*

CSS Essentials

overflow : controls how content that exceeds the size of its container is handled and displayed.

```
.box {  
  width: 200px;  
  height: 100px;  
  border: 1px solid black;  
  overflow: auto; /* Scrollbar appears only when needed */  
}
```

Main values :

- ***visible***: Content spills out of the container (default).
- ***hidden*** : Excess content is clipped and not visible.
- ***scroll*** : Scrollbars always shown, even if not needed.
- ***auto*** : Scrollbars shown only if content overflows.

CSS Essentials

position : property specifies how an element is positioned in the document.

Main values :

- ***static*** : Default. Element flows normally in the page (no positioning).
- ***relative*** : Positioned relative to its normal position; can be shifted using top, left, right, bottom.
- ***absolute*** : Positioned relative to the nearest positioned ancestor (not static). Removed from normal flow.
- ***fixed*** : Positioned relative to the viewport; stays fixed on screen during scroll. Removed from normal flow.
- ***sticky*** : Switches between relative and fixed depending on scroll position.

Thank you for attending
this presentation!